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Department of Computer Science and Engineering

Data Science

Diagnox - Explainable AI for Personalized Healthcare via multi modal fusion

BE_DS_A_Group No: 06

Name of The Guide:

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1. Problem Statement:

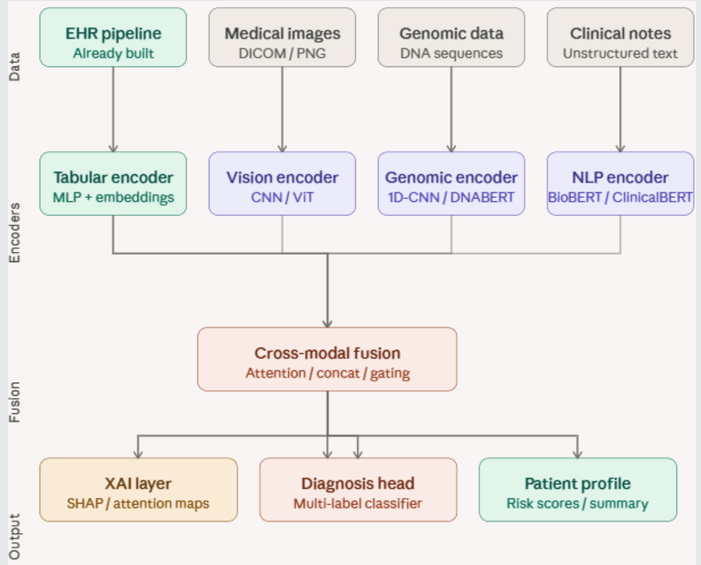
Current AI systems in healthcare are limited by fragmented data across EHRs, medical imaging, and genomics, preventing a unified patient view. Many models act as "black boxes," reducing clinician trust due to lack of explainability. They also fail to deliver personalized insights tailored to individual patients. Privacy and regulatory challenges further restrict real-world adoption.

DiagnoX addresses these gaps by integrating multimodal medical data and incorporating explainable AI to provide transparent, reliable, and patient-specific clinical decision support.

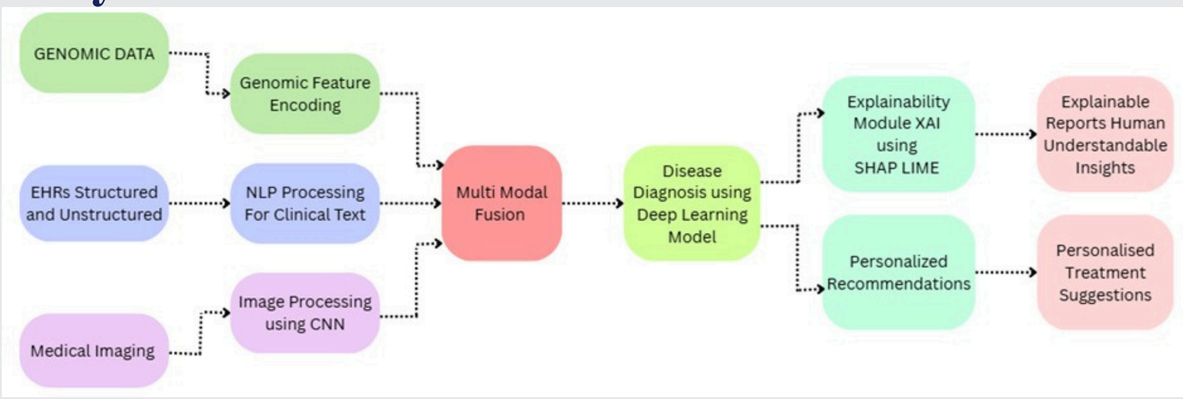
2. Objectives:

- Develop deep learning models to integrate EHRs, medical imaging, and genomic data for accurate and comprehensive diagnosis.
- Implement transformer-based NLP models to extract insights from unstructured EHR text for patient profiling and early detection.
- Apply advanced architectures to analyze medical images and genomic sequences for prescribe medicine.
- Incorporate explainable AI to provide transparent, interpretable predictions that build clinician trust and adoption.

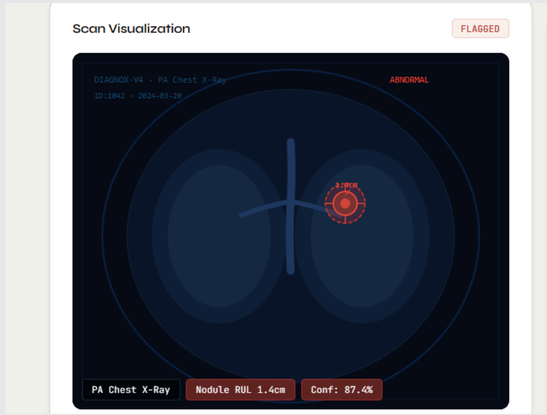
3. Workflow:

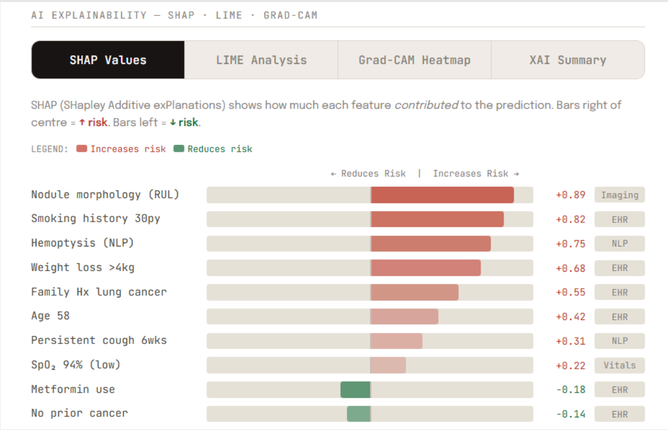


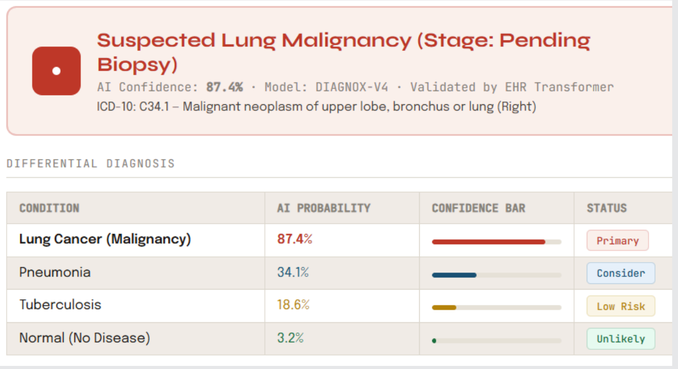
4. System Architecture:

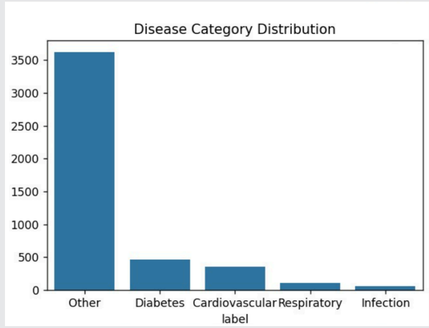


5. Results:









5. References:

[1] Heti Aprilianti, KhothibulUmam, and Maya Rini Handayani, *Comparative Study of SVM, KNN, and Naïve Bayes for Sentiment Analysis of Religious Application*

[2] Subhan Rizky Pratama and Ika Nur Fajri, *Comparison of K-Nearest Neighbor (KNN) and Support Vector Machine (SVM) Algorithms in Predicting Customer Satisfaction.*

[3] Agus Maula Rizki, Bustami, and Said Fadlan Anshari, *Comparison of Support Vector Machine and Naïve Bayes Algorithms in Sentiment Analysis*

6. Conclusion:

DiagnoX combines different medical data to help doctors find diseases more accurately and early. It also explains how the AI makes decisions, so doctors can trust it. Overall, it helps improve patient care in a simple and reliable way.